



POV PROPOSAL

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Agentic AI Proof of Value Proposal for Lead to Cash Management

1. Background and Objectives

Background — operating context

Rolls-Royce Power Systems (RRPS) runs an end-to-end lead-to-cash (L2C) process spanning customer engagement, product configuration, order creation, fulfilment, and invoicing. The application landscape is stable and mature with clear system responsibilities:

- CEC is the commercial system of record for opportunities, quotations, and dates. Incoterms and payment are defaulted from MS5 customer master.
- IPAS maintains configuration logic and bill-of-materials (BOM) used to translate a sold solution into manufacturable line items.
- MS5 (SAP ECC, SG instance) executes sales-order creation, order management, logistics, invoicing, and A/R.

These cores are supported by established integration and governance layers: SAP CPI is the sanctioned route for system-to-system connectivity into SAP; document repositories manage evidence and contractual artefacts; security and audit are enforced through tenant isolation, identity management, and change control.

Operational reviews highlight several persistent realities:

- Information captured upstream in CEC/IPAS is often re-keyed into MS5 to complete orders, lengthening cycle time and increasing inconsistency risk.
- Artefacts and policy dependencies that determine billing readiness (e.g., Incoterms clauses, approvals, FAT documentation) may sit outside the core transaction flow, making status hard to see at the moment it matters.
- Any improvement must stay within RRPS' posture: workloads in the RRPS tenant, tightly controlled network paths, and CPI as the single route into SAP.
- Stakeholders expect any evaluation to respect existing business flows (the “sell-solution” motion), avoid disruptive re-platforming, and produce auditable records that stand up to internal assurance.

In recent working sessions, RRPS and Integrum reviewed real user journeys, screen captures, and process notes to create a shared view of where friction appears, which artefacts are relied upon, which interfaces and schemas are available, and what technical/security constraints must be honoured. A joint action plan was formed to validate the as-is process, collate user guides and screenshots, trace data flows at screen/field level, review interface options, and lock the scope for a focused proof-of-value (POV). This plan anchors the proposal.

Objectives — what the POV will prove

The POV is not a **redesign**; it is a controlled evaluation conducted inside RRPS' guardrails. It will:

1. Quantify avoidable effort and delay in moving from “**order-ready**” to “**order posted**” using transaction-level evidence.

2. Establish baseline versus pilot outcomes for accuracy and first-time-right posting so leadership can judge whether targeted automation and verification materially improve quality (definitions in Section 4: Metrics).
3. Make billing readiness observable by demonstrating how required artefacts and policy checks can be evidenced, tracked, and audited alongside the transaction.
4. Prove compliance with RRPS' posture—data residency, security boundaries, and integration conventions—so any later scale-up, if approved, starts from a known-good pattern.

Scope

- In scope: tracing how data already present in CEC and IPAS is used to complete orders in MS5 for a small, representative set of order types; a measurement framework aligned to RRPS standards and ISO guidance that captures baselines, pilot results, and the associated audit trail (see Section 4).
- Out of scope: structural changes to master data, pricing design, or re-platforming of CEC, IPAS, or MS5.

2. Business Case

Purpose

Run a tightly scoped proof-of-value (POV) to determine whether RRPS can shorten the path from “order-ready” to “order posted,” raise first-time-right, and make billing readiness observable by reusing information that already exists in CEC and IPAS—without re-platforming or altering core business flows. Evidence will be gathered inside RRPS’ guardrails and assessed against the KPIs set out in Section 3: Metrics.

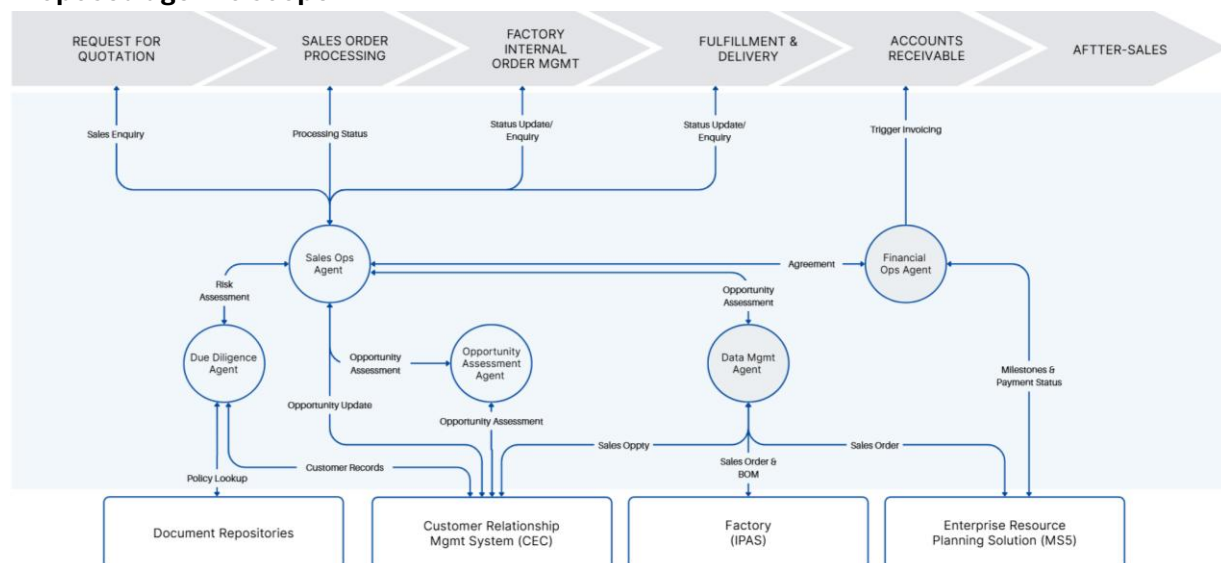
Value drivers (why this matters)

- Throughput & effort. Less swivel-chair re-keying and reconciliation during order creation; faster turnaround for in-scope order types.
- Quality & compliance. Commercial checks (partners, terms, Incoterms-driven artefacts) become explicit and auditable, reducing rework.
- Working capital. Earlier visibility of billing prerequisites (e.g., FAT evidence) reduces avoidable delays to invoice readiness.

What will change (agentic capabilities in scope)

- Sales Ops Agent (orchestrator). Assembles an order dossier from CEC and IPAS, proposes an explainable MS5 order (copy/lookup/derivation per field), coordinates checks and human approval, and posts via CPI with full provenance.
- Due Diligence Agent. Applies partner/terms/Incoterms policies; validates required artefacts (e.g., FAT); raises just-in-time tasks; stores an audit-ready DD summary; re-checks at billing milestones.
- Opportunity Assessment Agent. Scores late-stage opportunities for “order-ready,” flags gaps (partners, terms, dates, config health), based on data available from CEC.
- Data Mgmt Agent. Harmonises IPAS BOM → MS5 itemisation context; detects mismatches; serves canonical payloads/lookups to other agents.
- Financial Ops Agent. Monitors MS5 milestones/payment status; triggers compliant invoice steps when prerequisites are met; records control evidence.

Proposed agentic scope



Scope & boundaries

- Process span: CEC (late-stage opportunity) → IPAS (configuration/BOM) → MS5 (sales-order posting), with policy/document checks and billing-readiness prompts.
- Order coverage: a small, representative set agreed with Sales and Finance Operations.
- Guardrails: RRPS Azure tenant; SAP CPI as the sole SAP path; IPAS via existing proxy; no master-data or pricing redesign; no re-platforming.
- Out of scope: structural master-data changes, wholesale pricing model changes, and new customer channels.

Stakeholder impacts

Function	Today's friction	Improvement we expect	Evidence (see Section 4)
Sales / Sales Ops	Re-keying, late corrections	Explainable proposals; fewer edits; faster "ready→posted"	Auto-fill acceptance; cycle time
Finance Ops (A/R)	Artefact hunts; billing delays	Early DD tasks & evidence; fewer billing blocks	Billing-readiness observability; FTR
Supply Chain / Order Mgmt	Partner/terms errors, post-posting changes	Cleaner partners/terms; fewer corrections	FTR; exception trends
IT / Integration / Security	Pattern drift risk	Runs inside boundaries; immutable audit trail	Traceability coverage; change logs

Costs (POV level, opex-light)

- Configuration & mapping: field/rule mapping for selected order types; policy & artefact check catalogue; KPI instrumentation.
- Runtime & tooling: light Kubernetes namespace footprint in RRPS tenant; observability; identity/Key Vault; immutable audit store.
- Enablement & assurance: UAT with Sales/FinOps; operator walkthroughs; evidence-pack preparation.
- *No new platforms; CPI reuse wherever possible. Any new iFlows/OData surfaces will be narrowly scoped and change-controlled.*

Risks & controls

Risk	Impact	Control / Mitigation
Data gaps & edge cases	Lower auto-fill; more exceptions	Start with 1–2 order types; human-in-the-loop for uncertain derivations; feed exceptions into rule tuning
Interface constraints	Delivery slippage	Prefer CPI reuse; minimal net-new endpoints under change control; early interface inventory
Policy drift	Inconsistent outcomes	Versioned rule packs; change-managed prompts; DD summaries retained with orders
Adoption	Low acceptance of proposals	Explainability by default; one-click apply; reversible edits; short operator playbook
Privacy/compliance	Audit findings risk	Data minimisation; masked logs; no long-term LLM retention; DPIA/PIA checklist attached

Delivery approach & timeline

- **Weeks 1–2 — Discover & confirm:** validate as-is flow and rules; confirm interfaces; agree the KPI definitions and evaluation scenarios.
- **Weeks 3–6 — Configure & connect:** implement mappings, checks, and instrumentation; dry-runs on representative data.
- **Weeks 7–8 — UAT & refine:** user testing with Sales/FinOps; finalise evidence capture and runbooks.
- **Weeks 9–10 — Optional limited live cohort:** collect pilot results; compile the decision pack.

Acceptance at POV close

- **Operational uplift:** demonstrable reduction in manual effort and cycle time for in-scope orders.
- **Quality uplift:** higher first-time-right with auditable due-diligence evidence and artefacts.
- **Governance fit:** all activity proven to operate within RRPS' security and integration guardrails.
- **Evidence quality:** a board-ready pack with baselines vs. pilot outcomes, scenarios, data sources, and audit trail—enabling a clear scale/iterate/stop decision.

*(Architecture placement and SAP integration specifics follow in **Section 5: Architecture**; the end-to-end hand-offs appear in **Section 6: Data-Flow**.)*

3. POV Metrics

This scorecard compares a recent baseline with the pilot using identical definitions and data sources. It follows ISO/IEC TR 24030 practice: KPIs tied to objectives, clear data lineage, and scenario-based evaluation so results are comparable.

KPIs (definitions and owners)

KPI	Concise definition	Owner	Target band (to be discussed with business owners)
Field auto-fill acceptance	% of eligible MS5 header/item fields proposed by the agent and accepted by users	Sales Ops Agent	≥ XX%
First-time-right (FTR)	% of posted orders without rework due to data/policy issues	Due Diligence Agent	≥ +XX% uplift vs baseline
Cycle time (order-ready → posted)	Median minutes from “CEC/IPAS complete” to “MS5 posted”	Sales Ops Agent	≥ XX% faster
Billing-readiness observability	% of applicable orders with required artefacts (e.g., FAT) present before logistics/billing events	Due Diligence & FinOps	≥ XX%
Opportunity-readiness precision	% of opportunities marked “Ready” that post without DD gaps	Opportunity Assessment Agent	≥ XX%
Traceability coverage	% of pilot orders where every populated field has source (system/record/field) and CPI reference	Platform (cross-agent)	XX%

Use-case linkage and expected improvement

Use case (agentic scope)	Agents involved	KPIs applied	Expected improvement (POV band)	Benefits evidenced
Standard order (clean data) — late-stage CEC opportunity, valid IPAS BOM	Sales Ops, Due Diligence, Data Mgmt	Auto-fill, Cycle time, FTR, Traceability	Auto-fill 75–85%; Cycle time 30–40% faster; FTR +20–30%; Traceability 100%	Less re-keying; fewer corrections; audit-ready provenance
Configured order (rule-derived fields) — plant/sloc derived; one	Sales Ops, Data Mgmt, Due Diligence	Auto-fill, Cycle time, FTR, Traceability (+ BOM → Item check)	Auto-fill 60–75%; Cycle time 20–30% faster; FTR +20–25%; BOM → Item ≥98%;	Cleaner itemisation; fewer config clarifications

partner gap resolved			Traceability 100%	
Cross-border terms with artefact (e.g., FAT) — prompt, attach, verify	Due Diligence, FinOps, Sales Ops	Billing-readiness, FTR, Traceability	Billing-readiness $\geq 80\%$; FTR +20–25%; Traceability 100% (ArchiveLink/GO S ref)	Fewer billing blocks; earlier invoice triggers; clear control evidence
Upstream “Ready” gating — verify partners/terms/date/config health	Opportunity Assessment, Due Diligence	Readiness precision (+ downstream FTR/Cycle)	Readiness precision $\geq 85\%$; supports higher FTR and stable cycle in the above cases	Fewer late defects; smoother hand-off into order creation
Milestone-driven invoice triggers — observe MS5 milestones; trigger compliantly	FinOps, Due Diligence	Billing-readiness (trend)	Downward trend in time milestone → trigger; fewer billing blocks	Predictable billing cadence; improved working-capital rhythm
Spares visibility (read-only) — surface historical signals for after-sales	Supply Chain	Context metric (visibility)	Improved signal coverage on in-scope orders	Better transition into support; fewer follow-ups

Data and comparability

- **Sources:** CEC (opportunities/partners/terms/dates), IPAS (BOM/config lines), MS5 (orders, timestamps, rework/billing blocks), agent logs (suggest/accept with rule IDs), CPI message IDs, document-repository links/hashes.
- **Cohorts:** same pilot order types for baseline and pilot; comparable business weeks; any mix variance disclosed in the readout.
- **Controls:** correlation IDs across agent → CPI → SAP; immutable audit for the POV window; minimal PII with redacted logs; no long-term model retention.

Scorecard & readout

- **Columns:** KPI | Baseline | Pilot | Δ | Target band | Status (Green/Amber/Red).
- **Interpretation:** Green = at/above band (and Traceability = 100%) → candidate to scale; Amber = near band/variable → iterate; Red = below band or governance issues → fix or re-scope before scale.

Success statement

High acceptance of auto-filled fields, materially shorter order-ready-to-posted time, fewer rework events, and required artefacts present before billing events—each transaction fully traceable to source data and CPI calls.

4. Solution Feasibility

This section confirms that the POV can be delivered within RRPS guardrails and will behave reliably in day-to-day use. It covers functional behaviour by agent, the technical run-time and SAP integration, and a concise screen-to-field mapping with an automation coverage rationale.

Functional feasibility

- Opportunity Assessment Agent**
 Assesses whether a late-stage opportunity (based on the user defined stages) by checking partner completeness, terms/payment conditions, requested dates, and the state of configuration in IPAS.
- Sales Ops Agent (orchestrator)**
 Builds an order dossier from CEC (account/partners/terms/dates) and IPAS (configuration/BOM), calls Due Diligence, and MS5 order (each field labelled as copy/lookup/derivation). Supports user edits, then posts via SAP CPI. Captures the returned SAP document number and logs field-level provenance.
- Due Diligence Agent**
 Applies commercial and policy checks (partners/roles, Incoterms, approvals, pricing prerequisites). Where terms require artefacts (e.g., FAT), raises just-in-time prompts with a link to the repository, re-checks on completion, and stores an auditable DD summary attached to the transaction.
- Data Management Agent**
 Normalises IPAS BOM into a payload suitable for MS5 itemisation, validates material/quantity/UoM integrity, maps codes, and flags mismatches (e.g., unknown material, UoM conflict). Provides lookups to Sales Ops and FinOps agents.
- Financial Ops Agent**
 Observes MS5 milestones and payment status; when prerequisites are satisfied (including DD artefacts), raises invoice triggers in line with policy. Records the control evidence so A/R has a clear line of sight.

Technical feasibility

- Runtime placement**
 Agents run as stateless microservices on Kubernetes namespace in the RRPS Azure tenant. North-south access is via an Ingress/API Gateway (TLS termination, authN/authZ, rate-limits, audit logging). State services include PostgreSQL (operational data), Redis (caching), and object storage (artefact pointers/metadata).
- SAP connectivity (single path via CPI)**
 All SAP calls go Kubernetes namespace → SAP Integration Suite (CPI). Patterns:
 - CEC: OData (read) via CPI for opportunities, partners, terms, activities.
 - MS5 (ECC):
 - Simulate/validate via BAPI (synchronous) to keep UI responsive.
 - Create via IDoc ORDERS05 (asynchronous, durable, retryable).

- Attachments via CPI content adapter to GOS/ArchiveLink.
 - Events (delivery/invoice milestones) published by CPI to agent webhooks.
- **Security & privacy**
VNet isolation; IP allow-listing to CPI; least-privilege service principals; secrets in Azure Key Vault; TLS in transit; encrypted at rest. Logs are redacted; no long-term LLM retention. Each transaction carries a Correlation-ID and persists CPI Message IDs and SAP document numbers for replay and audit.
 - **Non-functional envelope (POV)**
99.5% agent API availability; proposal render $\leq \sim 3s$ p95 (excluding SAP round-trip); idempotent posting via correlation keys; exponential back-off on transient CPI/SAP errors; rate limits and throttling to protect backends. Central dashboards show request/error rates, latency, IDoc states, top business exceptions, and open DD tasks.

Screen-to-field mapping (summary)

This table shows representative mappings for the pilot order types. The Automation band reflects expected confidence during the POV.

MS5 field (illustrative)	Source & rule	Automation band	Notes
Sold-to (KUNNR)	CEC Account $\rightarrow$ validated to SAP code	Green	Direct copy after code lookup
Ship-to / Bill-to (KUNNR with PARVW = 'WE' (Ship-to), PARVW = 'RE' (Bill-to))	CEC partner roles (Ship-to/Bill-to)	Green	Fallback to default partner role rules if missing
Sales Org / Dist Ch / Division (VKORG/VTWEG/SPART)	From CEC opp/sales area; validate to allowed set	Green	Enforced via code table
Incoterms (INCO1/INCO2)	From CEC Terms; code-validate; INCO2 = location text	Green	Triggers DD artefact rules when applicable
Payment Terms (ZTERM)	From CEC Terms; code-validate	Green	Reject or prompt if not on allowed list
Customer PO No. (BSTNK/BSTKD)	From CEC quotation/opportunity	Green	Optional based on order type
Requested Delivery Date (EDATU)	From CEC; adjust by lead-time rule	Amber	Rule-derived; user can confirm/override
Plant (WERKS)	Derived (Ship-to geography $\times$ product line defaults)	Amber	Transparent rule; user confirmation
Storage Location (LGORT)	Derived (Plant $\times$ material class default)	Amber	Confirmable suggestion

Header/Item Texts (VBBK/VBBP)	From CEC notes + DD summary	Amber	Insert as long text with provenance
Material (MATNR)	From IPAS BOM line	Green	Direct copy after code validation
Quantity (KWMENG)	From IPAS BOM	Green	UoM normalised
Sales UoM (VRKME)	From IPAS UoM; convert if needed	Green	UoM map maintained by Data Mgmt
Attachments (FAT etc.)	DD provides link; CPI → ArchiveLink	Green	Stored externally; GOS reference captured

Automation coverage rationale

- Green (direct/lookup): master data keys, codes, BOM lines, UoM → high confidence auto-fill.
- Amber (rule-derived): plant, storage, requested dates, some texts → rule + user confirmation.
- Red (human-only): bespoke clauses, genuinely missing documents → prompts and tasks.

Based on the pilot order types, we expect $\geq \sim 70\%$ of the header/item fields to fall in Green, 15–25% in Amber, and the remainder in Red, aligning to the KPI targets in Chapter 3.

Exceptions & error handling (what happens when things go wrong)

- Business gaps (e.g., partner role missing, disallowed payment term): Due Diligence raises a task with a precise fix; Sales Ops re-checks and proceeds.
- Data mismatches (e.g., material not found, UoM conflict): Data Mgmt flags the line, offers a lookup/fix or marks the item red for manual completion.
- Policy conflicts (e.g., Incoterms require artefact not provided): create is blocked with a governed explanation; the task includes the correct repository path.
- Technical faults (CPI timeout, IDoc error): the platform retries with back-off; correlation keeps requests idempotent; the user sees a concise, actionable message and a link to the activity log.

Dependencies & readiness checklist (pre-build)

- Agreed pilot order type(s) and a final set of screenshots/user guides for mapping confirmation.
- Inventory of reusable CPI iFlows (OData/BAPI/IDoc) and any narrow new surfaces required.
- A short, versioned policy pack (Incoterms → artefacts, approval rules, allowed terms).
- Access and allow-lists: Kubernetes namespace → CPI egress ranges, OAuth2 client registration for CEC, CPI service users for ECC.
- Named SteerCo and business owners for UAT and acceptance.

5. Architecture & SAP Integration

The POV runs the agentic workload on Kubernetes namespace in the RRPS Azure tenant, with SAP Integration Suite (CPI) as the only route into SAP. Agents never call SAP directly. This keeps runtime inside RRPS' trust boundary, simplifies assurance, and maximises reuse of existing CPI assets.

Placement and trust boundaries

- Runtime: Agent services (Sales Ops, Due Diligence, Opportunity Assessment, Data Mgmt, FinOps, Supply Chain) and shared platform components run as containers on Kubernetes namespace.
- Ingress & API gateway: TLS termination, authentication/authorisation, rate-limiting, request logging.
- State services: PostgreSQL (operational data), Redis (cache), Object Storage (artefact pointers/metadata).
- SAP access: Kubernetes namespace → CPI over TLS with IP allow-listing from RRPS egress; CPI brokers to MS5 (ECC) and CEC.
- IPAS: reached via the existing proxy/firewall path; no new inbound exposure.

Integration model (simple, repeatable patterns)

- Read CEC: OData via CPI (opportunities, partners, terms, activities).
- Validate sales orders: BAPI via CPI (synchronous) for simulation/price/availability and error messages.
- Create sales orders: IDoc ORDERS05 via CPI (asynchronous, durable); idempotent retries handled by CPI.
- Attach evidence (e.g., FAT): CPI content adapter → GOS/ArchiveLink; only references are stored in the agent platform.
- Observe milestones: CPI publishes events/webhooks (delivery/invoice status) back to the agents.

Default flow: CEC OData read → BAPI simulate → IDoc create → ArchiveLink attach → events to agents.

Interface catalogue (initial POV scope)

- CEC (via CPI): GET Opportunity (expand Account/Partners/Terms), GET BusinessPartner, PATCH Opportunity (readiness note), GET Activities/Notes.
- MS5 / ECC (via CPI): BAPI_SALESORDER_SIMULATE, ORDERS05 inbound, optional ORDRSP outbound; ArchiveLink/GOS upload; milestone events.
- IPAS: read-only configuration/BOM through the existing proxy.

(CPI iFlows to be confirmed for reuse vs. net-new; changes will be narrowly scoped and change-controlled.)

Security controls

- Network: VNet isolation; CPI allow-listing; no inbound SAP exposure.
- Identity: OAuth2 client credentials for CEC OData; dedicated CPI service accounts for ECC; least privilege everywhere.

- Secrets: stored and rotated in Azure Key Vault.
- Transport & storage: TLS 1.2+; encryption at rest; logs redacted.
- Auditability: every transaction carries a Correlation-ID; CPI Message IDs and SAP document numbers (e.g., VBELN) are persisted with field-level provenance.

Canonical contracts (what we send/receive)

Agents exchange a **canonical JSON** with CPI; CPI maps to OData/BAPI/IDoc.

SalesOrderProposal (abridged)

```
{
  "header": {
    "salesOrg": "1000", "distChannel": "10", "division": "00",
    "soldTo": "CUST123", "shipTo": "CUST123-S", "billTo": "CUST123-B",
    "incoterms": {"code": "DAP", "location": "Singapore"},
    "paymentTerms": "0001", "requestedDate": "2025-09-30"
  },
  "items": [{"line": 10, "material": "MAT-001", "qty": 2, "uom": "EA", "plant": "SG01"}],
  "attachments": [{"type": "FAT", "uri": "rrps://docs/abc123", "hash": "sha256:..."}],
  "provenance": [{"field": "soldTo", "source": "CEC.Account.Id"}],
  "meta": {"correlationId": "klx-2025-000123", "dryRun": true}
}
```

- dryRun=true → CPI calls **BAPI simulate** and returns messages/pricing.
- dryRun=false → CPI converts to **ORDERS05** for **create** and returns the CPI Message ID; agents then record VBELN when available.

Error model (to users/ops)

```
{"correlationId": "klx-2025-000123", "status": "BUSINESS_ERROR",
  "issues": [{"code": "INCOTERMS_MISSING", "field": "header.incoterms.code", "severity": "ERROR"}]}
```

Business vs. technical faults are separated for clear action.

Sequencing (how a transaction flows)

Happy path

1. Sales Ops Agent assembles the dossier (CEC + IPAS).
2. Simulate via CPI/BAPI; show proposal with explanations; user approves.
3. Create via CPI/IDoc; persist Message ID, await/receive VBELN.
4. Due Diligence attaches evidence when required (ArchiveLink).
5. FinOps listens for milestones and raises compliant invoice triggers.

Common exceptions

- Missing partner/term → Due Diligence raises a task; re-check when resolved.
- Policy violation (e.g., disallowed payment term) → block with governed explanation.
- CPI/SAP transient error → retry with back-off; idempotent create via correlation keys.

Non-functional envelope (POV)

- Availability: ≥ 99.5% for agent APIs during POV.
- Performance: proposal render ≤ ~3s p95 (excluding SAP round-trip); create is asynchronous to keep UX responsive.

- Resilience: queued IDocs; exponential back-off; circuit breakers; rate-limits and throttling to protect SAP.
- Operability: dashboards for request/error rates, latency, IDoc states, top business exceptions, open DD tasks; audit export on request.

Environments and promotion

- DEV → QA → UAT → PROD with configuration-as-code for CPI iFlows and agent workflows.
- Test data: masked CEC/ECC records; synthetic artefacts for FAT.
- Cut-over: feature-flagged; run in “proposal-only” shadow mode before enabling posting.

Readiness checklist (to lock before build)

- Agreed pilot order types and screen artefacts for mapping confirmation.
- CPI inventory (OData, BAPI, IDoc) marked for reuse vs. creation.
- Policy pack (Incoterms → artefacts, approvals, allowed terms).
- Kubernetes namespace → CPI egress allow-lists; OAuth2 registration for CEC; CPI service users for ECC.
- Named owners for UAT and acceptance.

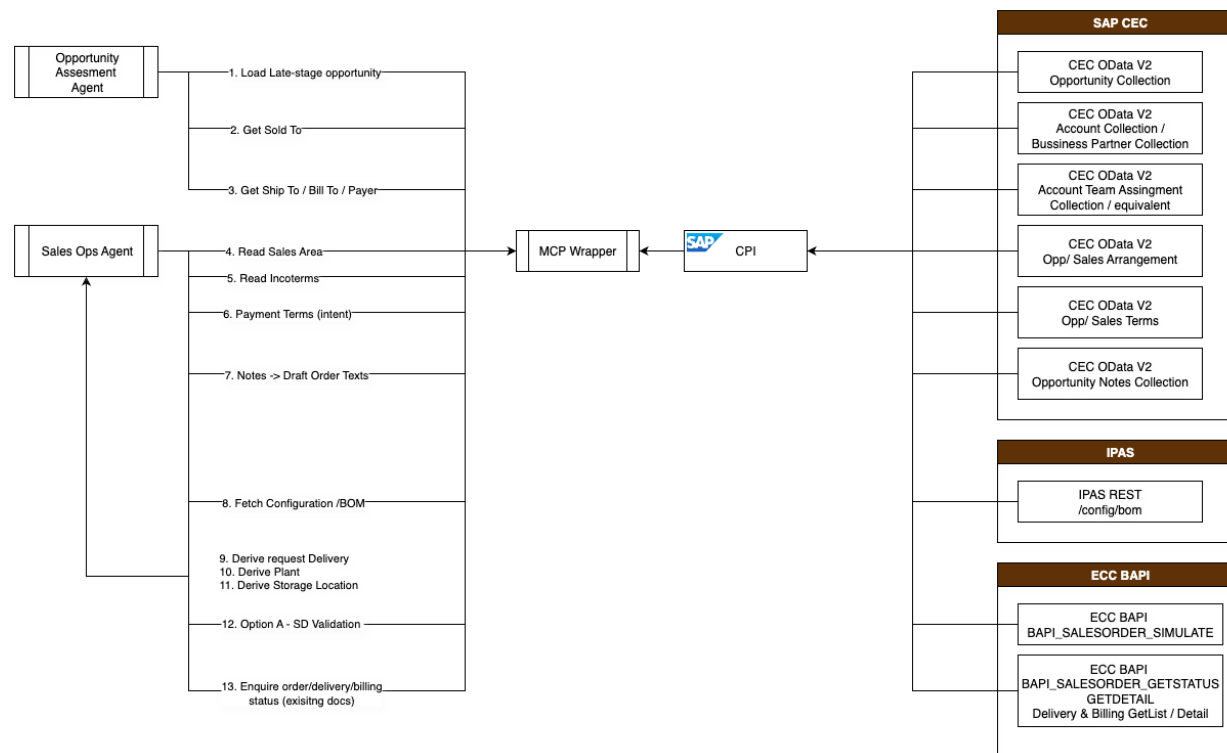
Outcome

This architecture delivers the POV inside RRPS’ security and integration posture: Kubernetes namespace-hosted agents, CPI-mediated SAP access, rule-driven explainable proposals, and end-to-end traceability. The next chapter describes the Data-Flow that these components execute from opportunity through to invoice readiness.

6 — Dataflow

This chapter provides a concise Input–Process–Output view of the lead-to-cash data path: commercial context originates in CEC, configuration comes from IPAS, the agentic layer on Kubernetes namespace consolidates and validates this as an order dossier, and all SAP transactions occur in MS5/ECC exclusively through SAP CPI. It sets the trust boundaries and ownership (Integrum for the agent payloads/orchestration; RRPS/CPI for IDoc iFlows and mappings), confirms that business approvals occur outside ECC, and specifies the interfaces and artefact handling at a high level (CPI-mediated reads/writes; ArchiveLink/GOS for documents). The section also defines the telemetry and audit anchors—correlation IDs, CPI message IDs, VBELN, and field-level provenance—so the flow is observable, governable, and aligned to the metrics in Chapter 4, without altering existing business processes.

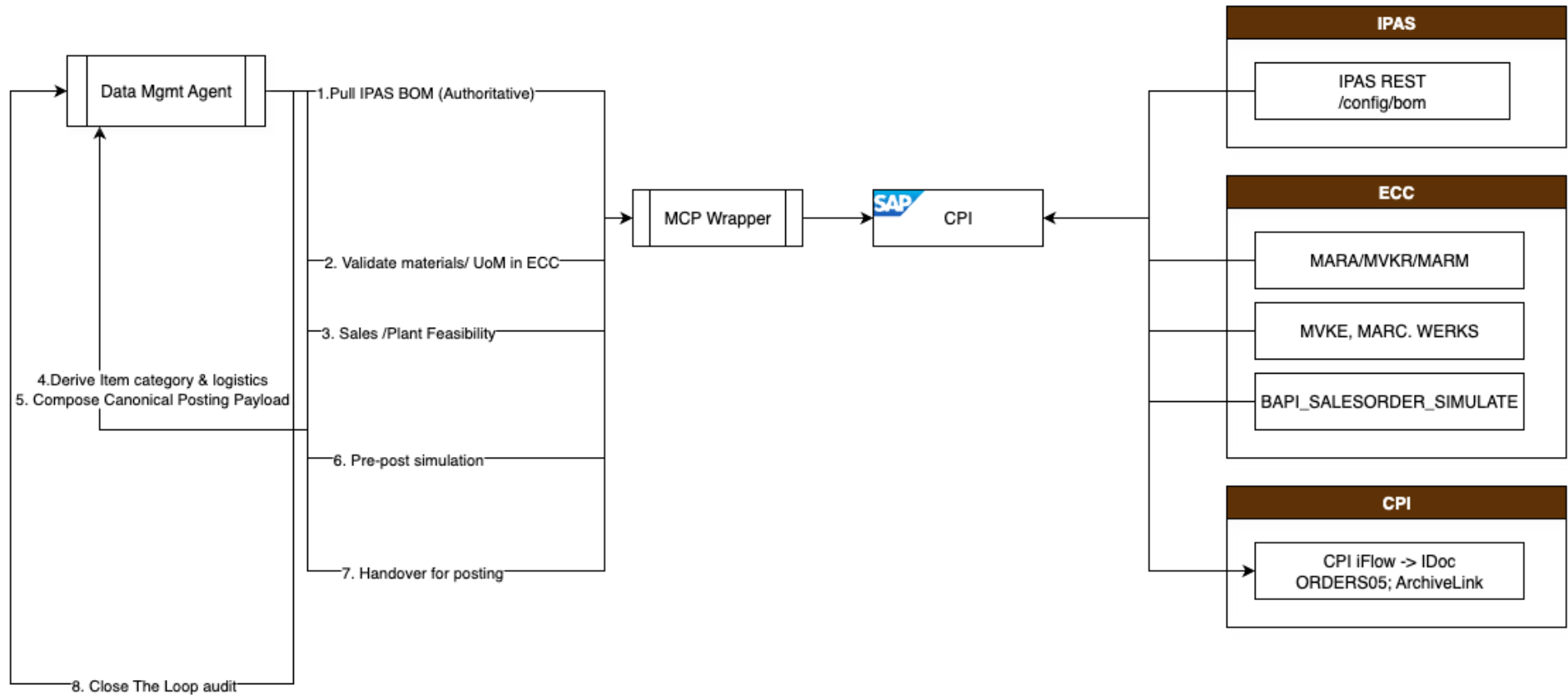
Sales Operations Support (Enquiry) - Input–Process–Output Flow



Step	Description	Actor	Source system & API	Data fields	Logic	Output
1	Load late-stage opportunity	Opportunity Assessment Agent	CEC OData v2 — OpportunityCollection	RequestedDate, ExternalReference, CustomerPO → VBEP-EDATU (sched preview), VBAK-BSTNK (hdr), VBKD-BSTKD (item, if used)	Validate presence; Derive schedule preview from requested date	Readiness context; gap list
2	Get Sold-to (Account)	Opportunity Assessment Agent	CEC OData — AccountCollection / BusinessPartnerCollection	Account/BP IDs → VBPA-KUNNR (PARVW='AG')	Copy + Validate (BP code lookup table)	Sold-to preview + provenance
3	Get Ship-to / Bill-to / Payer	Opportunity Assessment Agent	CEC OData — AccountTeamAssignmentCollection (or equivalent)	Role↔BP IDs → VBPA-KUNNR (WE/RE/RG)	Validate completeness per order type	Partner preview or actionable task list
4	Read Sales Area	Sales Ops Agent	CEC OData — Opp/Sales Arrangement	SalesOrg, DistributionChannel, Division → VBAK-VKORG/VTWEG/SPART	Copy + Validate against allowed set	Sales area confirmed in dossier
5	Read Incoterms	Sales Ops Agent	CEC OData — Opp/SalesTerms	IncotermsCode, IncotermsLocation → VBKD-INCO1/INCO2	Copy + Validate code; capture location text	Incoterms confirmed; DD note
6	Payment Terms (intent)	Sales Ops Agent	CEC OData — SalesTerms	Structured: PaymentTermsCode → VBKD-ZTERM (preview); Text-driven: PaymentTermsText → header VBBK (preview)	Validate code list; Template governed text when ECC uses free-text; no ECC write	Payment-terms validation + governed header text (preview)
7	Notes → draft order texts	Sales Ops / DD Agents	CEC OData — OpportunityNotesCollection	Notes → VBBK (header) / VBBP (item) (preview)	Template (limit length, add provenance)	Draft commercial & line texts (preview)
8	Fetch configuration/BOM	Sales Ops Agent	IPAS REST — /config/bom (proxy)	material, qty, uom → VBAP-MATNR/KWMENG/VRKME (preview)	Copy + Validate (UoM map)	BOM snapshot in dossier

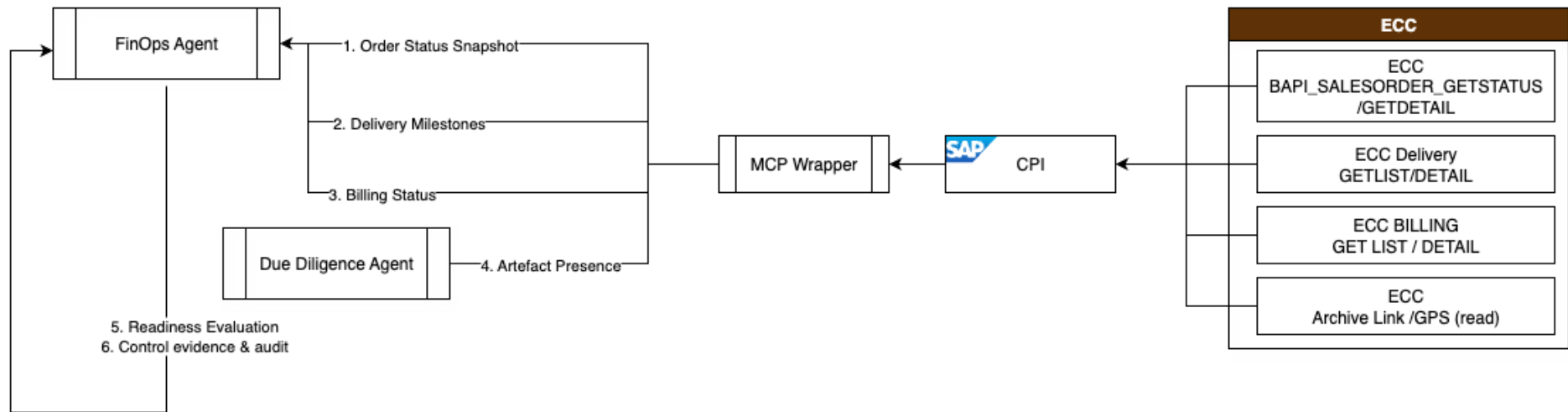
9	Derive requested delivery (sched)	Sales Ops Agent	From Step 1	Requested date → first schedule VBEP-EDATU (preview)	Derive (lead-time rules); flag out-of-tolerance	Proposed schedule date
10	Derive Plant	Sales Ops Agent	Platform rule (no external call)	Ship-to × product line → VBAP- WERKS (preview)	Derive; user- confirmable	Proposed plant + rationale
11	Derive Storage Location	Sales Ops Agent	Platform rule (no external call)	Plant × material class → VBAP- LGORT (preview)	Derive; user- confirmable	Proposed sloc + rationale
12	Option A — SD validation (single call)	Sales Ops Agent	ECC BAPI — BAPI_SALESORDER_SIMULATE	Sales area, partners, terms, items, schedule (previews above)	Query pricing/ATP/messages, no commit	Explainable proposal with SAP messages
13	Enquire order/delivery/billing status (existing docs)	Sales Ops / FinOps Agents	ECC — BAPI_SALESORDER_GETSTATUS/ GETDETAIL, Delivery & Billing GETLIST/DETAIL	VBUK/VBUP, LIKP/LIPS, VBRK/VBRP (read-only)	Query by VBELN/PO/Customer	Status panel consolidate d in dossier

Sales Operations Support (New Order) - Input-Process-Output Flow



Step	Description	Actor	Source system & API	Data fields (Source → SAP target)	Logic	Output
1	Pull IPAS BOM (authoritative)	Data Mgmt Agent	IPAS REST — /config/bom	material/qty/uom → VBAP-MATNR/KWMENG/VRKME	Copy; stage full BOM	BOM baseline
2	Validate materials/UoM in ECC	Data Mgmt Agent	ECC — master reads (MARA/MVKE/MARM)	MATNR existence; sales UoM; alt UoM	Query + Validate (code/UoM)	Exceptions list (unknown, UoM conflicts)
3	Sales/Plant feasibility	Data Mgmt Agent	ECC — MVKE, MARC-WERKS	Sales view present; plant availability	Query + Validate	Feasibility flags per item
4	Derive item category & logistics	Data Mgmt Agent	Platform rules	Material/type → VBAP-PSTYV; ship-to × product line → VBAP-WERKS; plant × material class → VBAP-LGORT	Derive; annotate rationale	MS5-ready itemisation (draft)
5	Compose canonical posting payload	Data Mgmt Agent	Platform assembly	Header/Partners/Terms/Items/Schedule → VBAK/VBPA/VBKD/VBAP/VBEP	Assemble; label Copy/Derive	Canonical JSON + mapping sheet
6	Pre-post simulation (optional)	Data Mgmt Agent	ECC BAPI — BAPI_SALESORDER_SIMULATE	Full payload as if to post	Query; collect pricing/ATP/error messages	“Clean-to-post” confirmation or fix list
7	Handover for posting (RRPS-owned)	RRPS CPI team	CPI iFlow → IDoc ORDERS05; ArchiveLink	Partners/items/terms/schedule; doc uploads	RRPS executes mappings & transports	Posted order VBELN; ArchiveLink IDs
8	Close-the-loop audit	Data Mgmt Agent	CPI/ECC responses	Correlate Message IDs, VBELN; persist provenance	Assemble	Posted order dossier + audit pack

Financial Operations Support (Billing Readiness) - Input-Process-Output Flow



Step	Description	Actor	Source system & API	Data fields	Logic	Output
1	Order status snapshot	FinOps Agent	ECC — BAPI_SALESORDER_GETSTATUS/GETDETAIL	VBUK (overall, delivery, billing, credit); VBUP (item)	Query by VBELN/PO/customer	Status tile in dashboard
2	Delivery milestones	FinOps Agent	ECC — Delivery GETLIST/DETAIL	LIKP/LIPS keys & status	Query by VBELN/ship-to/date	PGI/Delivery evidence timeline
3	Billing status	FinOps Agent	ECC — Billing GETLIST/DETAIL	VBRK/VBRP refs & status	Query by VBELN/period	Billing doc visibility
4	Artefact presence	Due Diligence Agent (supporting)	ECC ArchiveLink/GOS (read)	Linked docs on BUS2032 (key = VBAK-VBELN)	Query link tables; Validate type/hash	Evidence checklist (FAT/CoC/etc.)
5	Readiness evaluation (advisory)	FinOps Agent	Inputs: Steps 1–4	Terms VBKD-INCO1/INCO2; (read-only) block flags VBAK-FAKSK / VBAP-FAKSP	Evaluate prerequisites met?	Billing-readiness

						indicator + rationale
6	Control evidence & audit	FinOps Agent	Platform	Correlation-ID; CPI Message IDs; timestamps; user actions	Persist immutable log	Assurance pack for SteerCo/KPIs

7. Investment Summary

This 8-week POV is priced for transparency and tied directly to workstreams and deliverables.

Cost components

- **Professional Services (Implementation).** Design, build, and deployment of the Kubernetes namespace-hosted enquiry solution; sprint execution of agreed POV features.
- **AI Agent Enablement.** Configuration of Sales Ops, Due Diligence, Opportunity Assessment, and Financial Ops agents.
- **Integration Services.** Read-only connectivity via SAP CPI; CEC/IPAS data access; telemetry/provenance setup.
- **Testing & Demonstration.** End-to-end validation, UAT support, KPI snapshot, and POV Demonstration Report.
- **Cloud & Platform (if applicable).** Use of RRPS tenant/services; third-party licenses/subscriptions remain client borne.

Total Investment for POV: SGD 86,000

Less 10% discount

(A one-time 10% commercial consideration applied for the POV phase to support innovation partnership evaluation)

Total Investment for POV (Less Discount): SGD 77,400

Billing Milestones

The Proof of Value will be delivered by a lean, multidisciplinary team from Integrum over an eight-week Agile cycle, supported by designated RRPS stakeholders. The composition is designed to maintain velocity across functional validation, technical configuration, and user testing, while remaining cost-efficient and focused on value demonstration.

Team Composition

Role	Responsibilities
PM / Scrum Master	Leads the overall delivery cadence and ensures milestones are achieved within the agreed scope, timeline, and budget. Facilitates sprint planning and daily stand-ups, manages risks and dependencies, and maintains alignment between Integrum and Rolls-Royce teams. Consolidates progress updates, sprint reports, and prepares the SteerCo materials for review.
Solution Architect	Designs the end-to-end system architecture and data-flow patterns across CEC, IPAS, and MS5 via SAP CPI. Ensures the solution operates fully within Rolls-Royce's integration and security guardrails. Defines interface contracts, supervises environment setup, and validates that all agentic components maintain full auditability and traceability.

Proof of Value (PoV) Proposal

AI Business Consultant	Leads business discovery and process validation. Works closely with Sales Operations and Finance Operations to confirm the as-is and to-be processes, define policy and artefact requirements, and establish KPI measurement criteria. Guides UAT activities, interprets results, and ensures business outcomes are clearly demonstrated and documented.
AI Engineer	Develops, configures, and deploys the agentic AI components—Sales Ops, Due Diligence, Data Management, and FinOps agents. Manages API integration with CPI and other systems, performs test execution, tunes rule logic, and implements telemetry instrumentation. Supports UAT, resolves technical issues, and prepares the final demonstration environment.

Role	W1	W2	W3	W4	W5	W6	W7	W8	Total (FTE-weeks)	Approx. Days
PM / Scrum Master	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.3	2.0	10 days
Solution Architect	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2	2.1	11 days
AI Business Consultant	0.5	0.5	0.5	0.4	0.4	0.4	0.3	0.3	3.2	16 days
AI Engineer	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	8.0	40 days
Total Team Loading (FTE)	2.1	2.1	2.1	1.9	1.9	1.9	1.7	1.8	—	≈ 75 days

Summary

- Total Effort: ≈ 75 person-days (~1.9 FTE avg. over 8 weeks)
- Blended Rate: ≈ SGD 1 150 – 1 200 / day
- Total Investment: ≈ SGD 86 000 (before discount)

Professional Services Rate Card (for Change Requests)

Role	Rate (SGD / FTE / Day)
Project Manager / Scrum Master	1,100
Solution Architect	1,100
AI Business Consultant	1,000
AI Engineer	1,000
UI/UX Designer	750

Proof of Value (PoV) Proposal

Billing Milestones

Milestone	Timing	Linked Deliverables	Billing %
M1 – Initiation	Week 1	Project kick-off, access provisioned, backlog confirmed	5%
M2 – Core Enquiry Capabilities	End Week 3	Enquiry features live in dev; initial data integrations connected	10%
M3 – Operational Readiness	End Week 6	Consolidated views, UAT round completed, refinements applied	35%
M4 – Final Demonstration & Report	End Week 8	KPI snapshot, Demonstration Report, handover	50%

Payment Terms

- Currency: All fees are quoted and payable in Singapore Dollars.
- Invoicing: Invoices will be issued upon achievement of each billing milestone as set out above.
- Payment Period: Payment is due within 75 days of invoice date (Net 70) as per Rolls Royce procurement policy, unless otherwise agreed in the Master Services Agreement.
- Taxes: Fees are quoted exclusive of applicable taxes (e.g., VAT/GST). Any such taxes will be added to invoices where required by law.
- Expenses: Any pre-approved travel or incidental expenses incurred in support of this project will be billed at cost, supported by receipts.
- Overdue Payments: Late payments may be subject to interest at 1.5% per month or the maximum rate permitted by law, whichever is lower.

Assumptions & Risk Coverage

The successful delivery of the POV is based on the following assumptions, which also serve to mitigate delivery risks:

- Access to client tenant(s), SAP CPI read-only iFlows, and CEC/IPAS data is available for the duration of the POV.
- Scope is enquiry-only; no ECC postings, CPI IDoc mapping build, or ArchiveLink uploads.
- Client provides representative test data and SMEs for timely clarifications and UAT.
- Any change to scope, timelines, or access is handled via agreed Change Request.
- Success criteria will be validated with a pilot group of users.

To ensure clear accountability and efficient coordination across all involved parties, the following RICE matrix defines responsibilities at the activity and system level.

Proof of Value (PoV) Proposal

RICE Responsibility Matrix

Activity / Workstream	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Project governance, sprint cadence, reporting	Integrum PM / Scrum Master & RRPS PM	RRPS Project Sponsor	Integrum CEO	SteerCo
Architecture design and solution alignment	Integrum Solution Architect	RRPS Enterprise Architect	RRPS Business IT, Integrum PM	RRPS PM, SteerCo
Azure tenant setup, AKS deployment, VNet & Key Vault access	RRPS Cloud Infrastructure Lead	RRPS Head of IT Infrastructure	Integrum Solution Architect, RRPS PM	RRPS Security & Compliance
Integration validation via SAP CPI (OData/BAPI read-only)	Integrum Solution Architect & AI Engineer	RRPS CPI Owner	RRPS CPI Developer, RRPS Integration Governance, RRPS PM	RRPS IT Management
CEC data access (Opportunities, Partners, Terms)	Integrum AI Business Consultant	RRPS CEC Application Owner	Sales Ops Lead, RRPS Business IT	RRPS PM
IPAS configuration & BOM access	Integrum AI Engineer	RRPS IPAS Application Owner	RRPS Manufacturing Engineering SME	RRPS PM
MS5 (SAP ECC) read-only data extraction	Integrum AI Engineer	RRPS SAP Application Owner	RRPS Finance Ops, RRPS SAP Basis, RRPS PM	RRPS IT PMO
Data mapping and validation (CEC/IPAS → MS5)	Integrum AI Business Consultant	RRPS Business IT Lead	Sales Ops, Finance Ops SMEs, RRPS PM	SteerCo
Agent configuration & telemetry (Sales Ops, DD, Data Mgmt, FinOps)	Integrum AI Engineer	Integrum Solution Architect	RRPS CPI Team, RRPS Business IT, RRPS PM	SteerCo
Security review and environment compliance	RRPS Security Officer	RRPS CIO	Integrum Solution Architect, RRPS PM	SteerCo

Proof of Value (PoV) Proposal

User Acceptance Testing (UAT) & pilot validation	Integrum AI Business Consultant & RRPS PM	RRPS Business Process Owner	Sales Ops, Finance Ops SMEs	SteerCo
KPI measurement, evidence pack, and demo preparation	Integrum PM / Scrum Master & RRPS PM	RRPS Project Sponsor	Integrum Solution Architect, AI Business Consultant	RRPS Leadership
SteerCo presentation and closure report	Integrum PM / Scrum Master & RRPS PM	RRPS Project Sponsor	Integrum CEO, RRPS CIO	RRPS Leadership

Legend

- Responsible (R): Performs the work or ensures completion.
- Accountable (A): Owns the outcome and approves deliverables.
- Consulted (C): Provides expert input or domain validation.
- Informed (I): Receives updates or status communications.

This delineation ensures that ownership across business, IT, cloud, and integration layers is explicit, avoiding overlaps and ensuring the PoV operates within RRPS governance and security boundaries.

RRPS Resource Involvement Across the 8-Week PoV

The Proof of Value (PoV) will be delivered jointly by Integrum and RRPS teams, operating within RRPS' cloud and integration guardrails. RRPS resources will play defined roles throughout the 8-week delivery cycle, supporting environment readiness, data validation, testing, and user acceptance — while maintaining minimal disruption to business-as-usual operations.

The table below outlines the expected involvement and contribution by sprint.

RRPS Function / Role	Key Responsibilities	Sprint 0 – Infrastructure Setup (Weeks 1–2)	Sprint 1 – Opportunity Qualification (Weeks 3–4)	Sprint 2 – Order Data Retrieval & AI Validation (Core PoV)	Sprint 3 – SAP Submission & Audit Trail (Week 7)	Sprint 4 – UAT & Iteration (Week 8)	Estimated Effort (Days)	% of 8-Week Period
RRPS Project Manager	Leads internal project coordination, manages	●●●	●●●	●●●	●●	●●●	16	40 %

Proof of Value (PoV) Proposal

	dependencies, tracks milestones, supports SteerCo reporting							
CPI Integration Team	Validates existing iFlows, enables read-only OData/BAPI access, assists with interface mapping and test runs	●●●	●●	●●●	●●	●	8	20 %
Cloud Infrastructure / Azure Ops	Provides tenant access, VNet / Key Vault / AKS namespace setup, ensures network and connectivity integrity	●●●	●●	●	●	●	4	10 %
Business IT (CEC / IPAS / MS5)	Verifies field mappings, confirms data consistency, supports transformation and validation scenarios	●●●	●●●	●●●	●●	●●	10	25 %
Sales Ops / Finance Ops SMEs	Provides process walkthroughs, validates artefacts & policies, performs UAT and KPI validation	●●	●●●	●●●	●●●	●●●	10	25 %

Proof of Value (PoV) Proposal

Security & Compliance	Reviews connectivity , assesses data-handling controls, verifies PoV compliance within RRPS posture	●●●	●	●	●	●	4	10 %
SteerCo (Sponsor & Business Leads)	Participates in kick-off, mid-point review, and final demo for decision making	●●	●	●	●	●●●	2	5 %

Legend:

● = Light Involvement ●● = Moderate ●●● = Peak

Breakdown by Sprint

Sprint 0 – Infrastructure Setup (Weeks 1–2)

- Highest technical engagement across Cloud, CPI, and Security teams.
- Tenant access, VNet, Key Vault, and AKS namespace prepared under RRPS IT Infrastructure.
- RRPS PM coordinates readiness activities and confirms environment sign-off.

Sprint 1 – Opportunity Qualification (Weeks 3–4)

- Business IT and Sales / Finance SMEs validate opportunity data, readiness logic, and artefacts in CEC.
- CPI Team confirms interface connectivity for data retrieval.
- RRPS PM and Integrum PM manage joint sprint reviews.

Sprint 2 – Order Data Retrieval & AI Validation (Weeks 5–6)

- Core PoV feature development with active participation from Business IT, CPI, and SMEs.
- Data mapping (CEC → IPAS → MS5) validated; AI agents tested for order auto-fill accuracy.
- RRPS PM consolidates interim findings for governance review.

Sprint 3 – SAP Submission & Audit Trail (Week 7)

- CPI Team validates IDoc message flow and audit-trail capture.
- Security & Compliance ensures data handling within RRPS' integration and privacy standards.
- RRPS PM oversees milestone completion and prepares for UAT readiness.

Sprint 4 – UAT & Iteration (Week 8)

Proof of Value (PoV) Proposal

- Sales Ops and Finance SMEs execute user testing alongside Integrum's AI Business Consultant.
- RRPS PM reviews KPI metrics and evidentiary dashboards with Project Sponsor.
- SteerCo convenes to assess performance uplift and confirm PoV closure.

Summary

- Total RRPS Effort: $\approx$ 54 person-days ($\approx$ 1.3 FTE across 8 weeks).
- Peak Engagement: Infrastructure Setup and Core PoV (Sprints 0–2).
- Collaborative Governance: Continuous coordination between RRPS PM, Business IT, and Integrum PM.
- Minimal Disruption: Specialist teams (Cloud, CPI, Security) engaged in short, targeted windows; business users involved mainly during validation and UAT.

This structure ensures that RRPS oversight, data integrity, and compliance are maintained throughout the PoV while keeping operational load efficient and predictable.

8. Acceptance & Confirmation

We appreciate the opportunity to present this proposal and are available to discuss any aspect in greater detail to ensure alignment with your needs. Should you have any questions or require additional information, please feel free to contact Dr. Jack Hong at jack@integrum.global or +65 8023 8808.

To proceed, kindly review the proposal and confirm acceptance by signing below. Your confirmation will allow us to initiate the implementation process in accordance with the outlined scope, timeline, and investment structure.

Authorized Signatory:

Name:

Designation:

Company:

Date:

Authorized Signatory (Integrum Pte. Ltd):

Name: Dr. Jack HONG

Designation: CEO

Company: Integrum Pte. Ltd

Date: 12 Nov 2025

Appendix A — Beyond POV

1. Host Agentic Solution at Integrum

During the Proof of Value (POV) phase, the Agentic AI solution will be hosted within Integrum’s managed environment to provide a secure, controlled, and low-friction deployment experience.

This environment leverages the Kailash SDK runtime and its prefabricated modules (Dataflow, Nexus, Kaizen) to enable rapid activation, end-to-end orchestration, and continuous monitoring without requiring RRPS infrastructure dependencies.

Following successful validation, RRPS may choose to continue operations under Integrum’s managed environment for production-grade use or transition the runtime to its own tenant at a later stage.

Category	POV Phase (6 – 8 weeks)	Post-POV / Production Phase	Remarks
Hosting	Waived – covered by Integrum	SGD 2,000 – 3,000 per month (estimated)	Includes compute, storage, orchestration services, monitoring, and SOC 2-aligned security management within Integrum’s production cluster.
Token Consumption (LLM Processing)	Included under Integrum’s subscription	Chargeable to RRPS based on actual usage	- Estimated ~1.5 million tokens per month (≈ 100 sales orders). - Projected inference cost ≈ SGD 30 per month depending on model tier (GPT-4-turbo, Claude Sonnet, or Mistral Large). - Usage will be metered and reported monthly.
Maintenance & Support	Included in POV scope	22% of the implementation fees	Applies to post-POV managed service agreements only.

2. Deployment to RRPS environment

Following successful validation of the Agentic AI solution within Integrum’s managed environment, the full runtime—including all deployed agents, orchestration modules, and configuration assets—can be redeployed into RRPS’s own environment for extended testing or production enablement.

This deployment approach ensures data sovereignty, security compliance, and alignment with RRPS’s internal IT governance, while maintaining the same Kailash SDK architecture and orchestration pattern demonstrated during the POV.

Proof of Value (PoV) Proposal

The redeployment will leverage RRPS's existing cloud infrastructure (Azure or GCP tenant) and integration controls (e.g., SAP CPI, Key Vault, VNet isolation) to ensure seamless interoperability with internal systems.

Category	Description	Cost to RRPS	Remarks
Hosting	Runtime hosted and managed within RRPS's own environment.	<i>Not applicable</i>	RRPS will provide and maintain its own cloud infrastructure and related services.
Token Consumption (LLM Processing)	All LLM inference will be executed through RRPS's own enterprise subscription (e.g., Azure OpenAI, Anthropic, or Mistral).	<i>Borne by RRPS under its subscription</i>	Token usage will follow RRPS's internal billing and compliance processes. Integrum remains responsible for integration and monitoring readiness.
Redeployment & Configuration Services	Migration of agentic runtime, orchestration setup, and environment validation from Integrum's POV environment to RRPS's tenant.	Waived (complimentary)	Includes one-time migration, system validation, and first-run testing to ensure operational readiness.
Support & Maintenance	Ongoing enablement, performance monitoring, and agent updates.	22 % of implementation fees (per annum)	Covers technical support, version upgrades, configuration tuning, and governance dashboard maintenance.

3. Rollout agentic agents to similar BU

Following successful validation within RRPS Power Systems, the Agentic AI framework can be extended to additional business units (BUs) using the same Kailash SDK architecture and prefabricated modules.

This rollout leverages the reusable codebase, orchestration templates, and governance patterns established during the initial POV, allowing rapid enablement with minimal re-engineering.

The objective of this phase is to replicate the proven Agentic capability—such as Sales Ops, Due Diligence, and Financial Ops agents—across comparable business processes or regional entities, ensuring consistency, scalability, and cost efficiency.

Proof of Value (PoV) Proposal

Workstream	Description
Configuration Adaptation	Adaptation of existing agentic workflows and orchestration pipelines to the new BU's datasets, access roles, and operational context.
Integration Alignment	Re-mapping of connectors (e.g., CRM, ERP, telemetry feeds) where endpoints or policies differ. Core integration patterns remain unchanged.
Validation & UAT	Controlled testing with BU stakeholders to validate workflows, performance, and compliance.
Enablement & Training	Workshop sessions for BU operations and IT teams to manage runtime monitoring, configuration, and exception handling.

Assumptions:

- Core platform architecture, SDK runtime, and integration model remain consistent across BUs.
- Data structures, governance, and security frameworks follow RRPS global standards.
- RRPS provides access credentials, test data, and BU-specific process documentation.
- No material redesign or new system integration is required beyond configuration adaptation.

Component	Description	Estimated Effort	Indicative Cost (SGD)
Professional Services	Configuration, validation & enablement for one additional BU	~ 3 – 4 weeks	SGD 39 000
Hosting	Incremental compute, storage & monitoring overhead for additional BU runtime under shared Integrum environment	—	SGD 600 per month (≈ 7 200 per annum)
Support & Maintenance	Annual technical support, updates & performance monitoring for the extended BU runtime	—	22 % of implementation = SGD 8 580 per annum
Token Consumption	Option A – RRPS Enterprise Subscription: LLM inference executed under RRPS's own LLM tenant (e.g., Azure OpenAI, Anthropic, Mistral). Costs billed directly to RRPS. Option B – Integrum-Managed	—	Option A: Billed directly by RRPS LLM provider Option B: Billed by Integrum at the same rate with monthly usage

Proof of Value (PoV) Proposal

	Subscription: Inference hosted via Integrum’s enterprise account; usage metered and billed by Integrum based on actual token consumption.		reporting (≈ SGD 30 / month for 100 orders).
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Notes:

- The proposed rollout model follows enterprise reuse principles, ensuring that each new BU instance inherits the validated SDK runtime, orchestration patterns, and governance framework without duplication of build cost.
- Incremental hosting cost reflects the marginal compute, storage, and observability overhead expected when multiple BUs operate concurrently within the shared managed environment.
- Token-consumption options are provided to preserve commercial flexibility and data-sovereignty alignment with RRPS’s internal IT policy—RRPS may switch between self-managed and Integrum-managed inference at any time without architectural change.
- The overall structure ensures that scalability is linear, transparent, and operationally sustainable, allowing RRPS to expand adoption across multiple business units while maintaining predictable total cost of ownership (TCO).
- Support and maintenance at 22 % per annum provides lifecycle assurance, security patching, version alignment, and proactive monitoring consistent with Integrum’s enterprise SLA framework.

4. Extend agentic agents to cover different scope

Beyond the current implementation, RRPS may progressively extend the Agentic AI capability within the existing Lead-to-Cash (L2C) framework.

Such extensions may include adding new functional logic to existing agents or developing additional agent modules to support related business workflows, analytics, or governance needs.

All extensions will continue to leverage the validated Kailash SDK runtime and prefabricated modules (Dataflow, Nexus, Kaizen), ensuring full compatibility with the established architecture, integration model, and security posture.

Scope of Work

Workstream	Description
Requirements & Design	Joint definition of new functional scope or agent capability within the existing runtime.
Configuration & Development	Implementation of new logic, orchestration nodes, or workflow steps using the Kailash SDK.
Integration & Validation	Testing and validation to ensure compatibility with current agents, datasets, and SAP CPI interfaces.

Proof of Value (PoV) Proposal

Enablement & Documentation	Updating governance playbooks, audit dashboards, and operational runbooks for the expanded scope.
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Assumptions:

- The existing SDK runtime and integration architecture remain unchanged.
- Enhancements are delivered within the validated Lead-to-Cash environment.
- RRPS provides access credentials, data fields, and SME participation for UAT.
- Work is performed on a **time-and-materials basis**, following the agreed rate card.

Commercial Model

Component	Basis	Effort / Rate	Indicative Costing Approach
Professional Services	Design, configuration, validation & enablement of new or enhanced agent functionality	<i>Time & Materials</i>	Based on rate card: • Project Manager / Scrum Master – SGD 1 100 / day • Solution Architect – SGD 1 100 / day • AI Business Consultant – SGD 1 000 / day • AI Engineer – SGD 1 000 / day • UI/UX Designer – SGD 750 / day
Hosting	Incremental compute, storage & monitoring for the expanded runtime under Integrum-managed environment	—	SGD 400 / month ($\approx$ 4 800 per annum) ¹
Support & Maintenance	Ongoing support, version updates & performance monitoring for the extended scope	—	22 % of actual implementation fees (T&M total) per annum
Token Consumption	Option A – RRPS's own enterprise LLM subscription (billed directly to RRPS). Option B – Integrum-managed inference (metered & billed monthly).	—	Usage-based; cost proportional to token volume

Proof of Value (PoV) Proposal

¹ Hosting fee applies only where the additional functions are operated under Integrum's managed environment. No fee applies if RRPS self-hosts within its own tenant.

All fees quoted in Singapore Dollars (SGD), exclusive of GST/VAT.

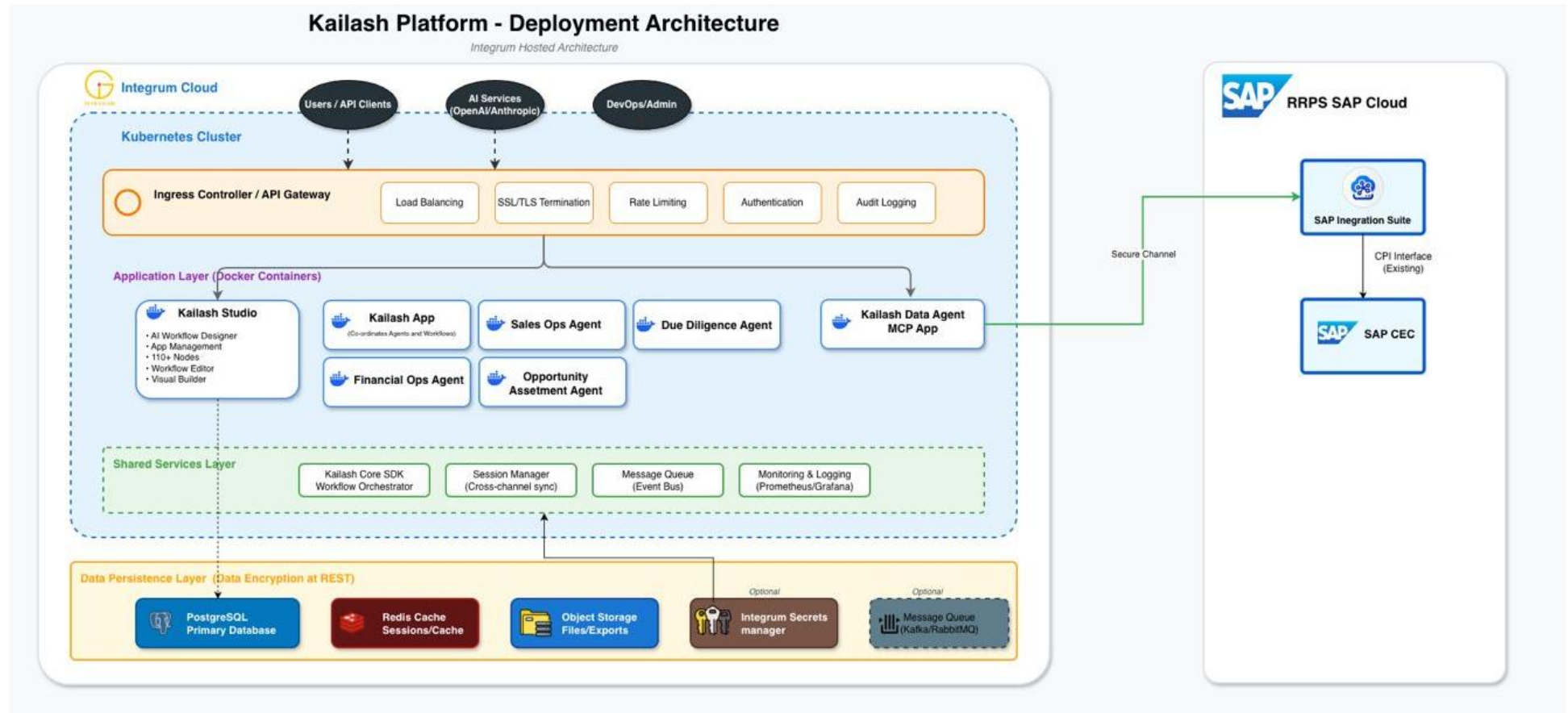
Remarks

- Time-and-materials pricing provides RRPS with flexibility to scale enhancement work according to business priorities and available budget.
- Work estimates will be scoped and approved jointly prior to commencement; detailed time-sheets and sprint reports will be provided for transparency.
- Support & Maintenance remain at 22 % of the actual implementation total, ensuring lifecycle coverage and alignment with enterprise SLA standards.
- Hosting reflects incremental compute overhead only; no new infrastructure provisioning is required.
- Token usage continues under RRPS's selected model (self-managed or Integrum-managed), maintaining clear cost attribution.
- This model enables RRPS to incrementally expand agentic intelligence and automation within the same architecture—adding measurable capability while preserving governance, observability, and predictable cost control.

Appendix B — Glossary

Term	Meaning
Agentic AI	A set of cooperating software agents that read context, propose actions with explanations, and execute with human-in-loop control.
Sales Ops Agent	Orchestrates dossier build, simulation, proposal, approval, and order creation via CPI.
Due Diligence Agent	Checks partners/terms/Incoterms, manages artefact tasks (e.g., FAT), and records control evidence.
Opportunity Assessment Agent	Assesses “order-ready” at late-stage opportunities.
Financial Ops Agent	Observes milestones and prerequisites, recommending/clearing billing blocks when evidence is present.
Data Mgmt Agent	Normalises IPAS BOM into MS5 itemisation and validates codes/UoM.
CEC	Customer engagement/CRM system (opportunities, partners, terms, activities).
IPAS	Product configuration and BOM system used to derive manufacturable line items.
MS5 (ECC)	SAP ECC (SG instance) used for order creation, fulfilment, invoicing, and A/R.
CPI	SAP Integration Suite; the only path into SAP from Kubernetes namespace (simulate = BAPI; create = IDoc).
BAPI Simulate	Synchronous sales-order simulation used to validate pricing/availability and catch business errors.
IDoc ORDERS05	Asynchronous, durable interface used to create sales orders in ECC.
ArchiveLink / GOS	SAP attachment framework; documents are stored against the sales order (object key = VBELN).
VBELN	SAP Sales Document Number returned after successful posting.
FAT	Factory Acceptance Test documentation (example of a billing prerequisite).
FTR	First-time-right posting (posted without data/policy rework).
Order dossier	The assembled set of header/items/terms/partners + provenance from CEC/IPAS that drives the proposal.

Appendix C — Proposed Architecture



Appendix D — Sprint Timeline and User Stories

Sprint	Duration	Epic / User Stories	User Story Narrative
Sprint 0 – Infrastructure Setup	Nov 17 – 28, 2025	<i>Environment Setup & Validation</i>	Establish the foundational environment, validate the Kailash SDK and CPI integrations, and build the notification infrastructure. This ensures that all agents can operate securely in a controlled, read-only environment across C4C, IPAS, and MS5.
Sprint 1 – Opportunity Enquiry Agent	Dec 1 – 12, 2025	Story 1.1: Enquire on Customer Profile	As a Sales Manager, I want to be able to make enquiries on a specific customer profile, so that I can view their summary, lifetime value, and activities to better assess sales potential. Acceptance Criteria • Sales Manager can type a query in chat (e.g., “Show me ACME’s profile”). • System responds in chat with structured summary (customer profile, lifetime value, activities). • Data is retrieved from MS5 (orders/financials), IPAS (technical data), and CEC (prospects/customers/opportunities). • For new prospect, system should validate the RR sanction and blacklist database lookup Design / System Enhancements • Extend enquiry service to support conversational queries. • Responses structured as chat messages with clear sections (Profile Summary, Lifetime Value, Activities). • Chat UI allows Sales Manager to refine requests (e.g., “filter last 12 months”). Testing & Validation • Unit tests for data retrieval from MS5, IPAS, and CEC. • Integration tests validating correct aggregation into a single chat response. • UAT with Sales Managers to confirm natural query-response flow.

			Technical Approach • Build API connectors to MS5, IPAS, and CEC. • Implement consolidated endpoint /api/chat/customer/profile/:id. • Integrate chat NLU to interpret profile-related queries. • Format chat response into structured sections (summary card-style inside chat). Subtasks • Build backend connector for MS5. • Build backend connector for IPAS. • Build backend connector for CEC. • Implement profile aggregation logic. • Integrate structured output into chat interface. Dependencies • Access to MS5, IPAS, and CEC. • Chat framework with natural language query support. Risk Assessment • Medium: Natural language queries may be ambiguous → mitigation: provide clarifying prompts. • Low: Large profiles may clutter chat → mitigation: show concise summary with “expand” option.
		Story 1.2: Review Purchase History & Projects	As a Sales Manager, I want to enquire on a customer’s purchase history and projects, so that I can understand past behaviour and identify future opportunity areas. Acceptance Criteria • Chat interface allows a Sales Manager to request a customer’s purchase history. • System responds in chat with summarized purchase records (order values, dates, products).

			- Projects associated with purchases are listed in context with the orders. - Sales Manager can refine queries via chat (e.g., filter by date, product, or project). Design / System Enhancements - Extend enquiry service to retrieve and summarize purchase history + project data. - Integrate natural language understanding so chat queries like “<i>Show me ACME’s last 3 projects</i>” return structured results. - Responses formatted as structured chat messages with expandable details. Testing & Validation - Unit tests for MS5 order retrieval and IPAS project data. - Integration tests for chat-based retrieval and display. - UAT validation with Sales Managers testing natural query flows. Technical Approach - API connector to MS5 for purchase history. - API connector to IPAS for project data. - Aggregated endpoint /api/chat/customer/history/:id. - Chat interface returns results as text + structured cards within the chat window. Subtasks - Build backend connector for MS5 purchase history. - Build backend connector for IPAS project data. - Implement data aggregation service for chat responses. - Integrate natural language parsing for queries. - Format chat output (summary with expandable purchase/project details). Dependencies - Access to MS5 (order history). - Access to IPAS (projects). - Chat interface framework with NLU support. Risk Assessment
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			• Medium: Ambiguity in natural language queries (mitigation: define fallback prompts and clarifying questions). • Low: Large histories may clutter chat (mitigation: provide summarized views with “show more” option).
		Story 1.3: Access Product Information & Insights	As a Sales Manager, I want to enquire on products linked to the customer and access the latest external insights, so that I can have up-to-date information for opportunity engagement. Acceptance Criteria • Sales Manager can type a query in chat (e.g., “<i>Show me products ACME has purchased</i>” or “<i>Any recent news on ACME’s product line?</i>”). • System responds with a structured list of products associated with the customer. • System retrieves and summarizes external insights/news relevant to those products or the customer. • Insights are attributed with source references (e.g., article title, link). Design / System Enhancements • Extend enquiry service to pull product information from MS5. • Integrate external news feeds/APIs for market and customer insights. • Format chat response as a structured message with two sections: Products and Latest Insights. Testing & Validation • Unit tests for product retrieval from MS5. • Integration tests for combining product data with external news sources. • UAT validation with Sales Managers to confirm clarity and usefulness of insights. Technical Approach • API connector to MS5 for product data. • Integration with external news API for insights.

			- Endpoint /api/chat/customer/products/:id delivering products + insights. - Chat response formatted with expandable product list and news snippets. Subtasks - Build backend connector for MS5 product data. - Integrate external news API. - Implement aggregation logic for products + insights. - Format results as structured chat response. - Add attribution (links/sources) to insights in chat output. Dependencies - Access to MS5 (product data). - External news feeds/APIs. - Chat interface framework with support for rich messages (links, expandable lists). Risk Assessment - Medium: External insights may return irrelevant or noisy results → mitigation: apply keyword filtering or relevance scoring. - Low: Product list may be too long for chat → mitigation: show top items with option to “load more.”
Sprint 2 – Order Data Enquiry & Sales Operations Notification	Dec 15 – 26, 2025	Story 2.1: Receive Product & Configuration Suggestions	As a Sales Operations Specialist, I want to receive AI-driven suggestions on production specifications and configurations in IPAS through a web-based chat interface, so that I can propose accurate solutions before the order booking stage. Acceptance Criteria - Sales Ops Specialist can request configuration guidance in chat (e.g., “Suggest configuration for ACME’s last order”). - System analyses customer-specific history from MS5 and IPAS. - Suggestions include recommended specs, configurations, or adjustments. - Output is displayed in chat as structured recommendations with rationale. Design / System Enhancements

			• Extend enquiry service to analyse customer history and generate tailored configuration suggestions. • Chat responses formatted with Suggested Configurations section and supporting context. • Allow follow-up refinements in chat (e.g., “<i>filter for renewable options</i>”). Testing & Validation • Unit tests for retrieval of customer-specific history. • Integration tests validating accuracy of configuration suggestions. • UAT with Sales Ops Specialists to confirm usefulness and clarity. Technical Approach • API connector to IPAS for configuration data. • Access MS5 for historical order patterns. • ML/AI logic layer to generate tailored configuration suggestions. • Endpoint /api/chat/order/suggestions/:id. • Responses formatted as structured recommendation cards inside chat. Subtasks • Build backend connector for IPAS configuration data. • Extend MS5 connector for historical usage patterns. • Implement AI-driven suggestion engine. • Format chat output with structured recommendations. • Enable follow-up refinements in chat queries. Dependencies • Access to MS5 (order history). • Access to IPAS (production configuration data). • Chat interface framework with support for structured recommendation output. Risk Assessment • Medium: AI suggestions may be too generic → mitigation: train with customer-specific history and feedback loop. • Low: Complex recommendations may not fit chat format → mitigation: provide summary with “view details” option.
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		Story 2.2: Consolidate Order Details	As a Sales Operations Specialist, I want to consolidate order-related information from CEC (opportunity) and IPAS (product configuration) into one view through a web-based chat interface, so that I can validate completeness before creating the order in MS5. Acceptance Criteria • Sales Ops Specialist can query in chat (e.g., “<i>Show me ACME’s order details before booking</i>”). • System retrieves opportunity data from CEC and configuration data from IPAS. • Data is combined into a structured response in chat. • Response highlights any missing or incomplete fields before MS5 order creation. Design / System Enhancements • Enquiry service aggregates opportunity and configuration data. • Chat interface delivers structured response with sections for Opportunity (CEC) and Configuration (IPAS). • Validation rules flag missing values or inconsistencies. Testing & Validation • Unit tests for CEC and IPAS data retrieval. • Integration tests for combined order details view. • UAT validation with Sales Ops Specialists simulating real opportunities. Technical Approach • API connector to CEC for opportunity data. • API connector to IPAS for configuration data. • Consolidated endpoint <code>/api/chat/order/details/:id</code>. • Chat response structured into sections: <i>Opportunity Details</i> and <i>Configuration Details</i>, with validation flags. Subtasks • Build backend connector for CEC opportunity data.
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			• Build backend connector for IPAS configuration data. • Implement aggregation logic for combined view. • Add validation checks for completeness. • Format chat output into structured sections with flags. Dependencies • Access to CEC (opportunity data). • Access to IPAS (product configurations). • Chat interface framework for structured responses. Risk Assessment • Medium: Inconsistencies between CEC and IPAS records → mitigation: reconciliation logic with priority rules. • Low: Chat response could become too detailed → mitigation: allow summarized output with “expand details.”
Sprint 3 – Sales Operations & Sales Manager Notifications	Dec 29 – Jan 9, 2026	Story 3.1: Generate Entry Template from PO	As a Sales Manager, I want the system to generate an entry template in MS5 upon receipt of a customer’s purchase order through a web-based chat interface, so that I can ensure accurate and consistent order booking. Acceptance Criteria • Sales Manager can upload or reference a PO in the chat interface. • System automatically extracts PO details (e.g., order items, quantities, pricing). • An MS5 entry template is generated and presented back in chat for review. • Any missing or inconsistent data is flagged in the chat response. Design / System Enhancements • Chat interface supports PO file upload or text entry. • PO parsing service extracts key data fields. • MS5 entry template auto-populated with extracted details. • Chat response provides a link or preview of the generated entry template. Testing & Validation • Unit tests for PO parsing logic.

			• Integration tests to verify MS5 template generation. • UAT validation with Sales Managers to confirm smooth PO-to-template flow. Technical Approach • Implement PO ingestion module (supports PDF/Excel/text formats). • Data extraction engine maps PO fields to MS5 fields. • Endpoint /api/chat/order/template/:id returns entry template reference. • Chat interface provides confirmation message and “review before submission” option. Subtasks • Enable PO upload within chat interface. • Implement PO data extraction service. • Map extracted data to MS5 entry fields. • Integrate validation and error-checking. • Deliver chat response with template preview. Dependencies • MS5 system for order booking. • File parsing libraries for PO formats. • Chat framework with upload support. Risk Assessment • Medium: Parsing errors from unstructured POs → mitigation: support standard formats and add manual correction flow. • Low: Integration with MS5 may require custom API adapters → mitigation: align mapping logic with MS5 schema early.
		Story 3.2: View Order Status & History	As a Sales Manager, I want to enquire on the status and history of any order through a web-based chat interface, so that I can track progress, see when updates occurred, and follow up with customers efficiently. Acceptance Criteria

			• Sales Manager can request status in chat (e.g., “Show me status of Order #12345”). • System responds with current order status (e.g., confirmed, in production, shipped). • Chat response includes key timestamps and history of changes (created, updated, released). • Notifications are triggered in chat when an order is updated. Design / System Enhancements • Order enquiry service fetches real-time status and logs from MS5. • Chat response formatted as a timeline: <i>Created</i> → <i>Updated</i> → <i>Released</i> → <i>Current Status</i>. • Notifications integrated into chat for proactive updates. Testing & Validation • Unit tests for order retrieval from MS5. • Integration tests for timeline accuracy across updates. • UAT validation with Sales Managers using real sample orders. Technical Approach • API connector to MS5 for order status and audit history. • Endpoint /api/chat/order/status/:id returns structured timeline. • Push notification integration with chat for real-time updates. • Timeline formatted as a structured message in chat. Subtasks • Build connector for MS5 order status/history. • Implement chat response formatter for timeline view. • Enable proactive notifications in chat when order status changes. • Test with live MS5 orders. Dependencies • MS5 for order and status history. • Chat interface framework with support for structured timeline messages and notifications.
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			Risk Assessment • Medium: Notification overload for Sales Managers → mitigation: allow subscription per order. • Low: Data sync delays between MS5 and chat interface → mitigation: implement retry and caching logic.
Sprint 4 – Order Management Operations & UAT	Jan 12 – 23, 2026	Story 4.1: Validate Order & Initiate Billing	As a Finance Staff, I want to validate order details and initiate billing directly from the chat interface, so that I can confirm accuracy and trigger invoices without leaving the workflow. Acceptance Criteria • Finance Staff can input a command (e.g., “<i>Validate Order #12345 for billing</i>”). • System retrieves full order details from MS5 (customer info, order items, amount). • Finance Staff can confirm or reject order for billing via chat. • If confirmed, billing process is initiated in MS5. • Confirmation message displayed in chat: “<i>Order #12345 validated and billing initiated.</i>” Design / System Enhancements • Chat interaction supports validation flow: 1. Display order details. 2. Ask for confirmation. 3. Trigger billing upon confirmation. • Errors or mismatches flagged in chat (e.g., missing data, incorrect amounts). • Audit trail maintained in MS5 for each validation. Testing & Validation • Unit tests for fetching order data from MS5. • Integration tests for validation and billing trigger flow. • UAT with Finance Staff to confirm usability and accuracy of chat validation. Technical Approach

			• API connector to MS5 for order retrieval and billing trigger. • Chat workflow with state handling (validate → confirm → trigger). • Endpoint /api/chat/order/validate/:id returns order details for confirmation. • Endpoint /api/chat/order/bill/:id initiates billing. Subtasks • Build connector to fetch order details from MS5. • Implement billing initiation API call to MS5. • Design chat workflow for validation and confirmation. • Add audit logging for Finance validations. • Conduct Finance-focused UAT session. Dependencies • MS5 (Order management and billing APIs). • Chat interface with multi-step conversational support. Risk Assessment • High: Incorrect billing if validation flow is bypassed → mitigation: enforce confirmation step in chat. • Medium: Integration errors with MS5 → mitigation: fallback error message in chat and retry mechanism. • Low: Finance Staff resistance to chat workflow → mitigation: provide training and step-by-step guides.
		Story 4.2: Manage Down Payment & Balance Monitoring	As a Finance Operations Specialist, I want to track issuance of down payments based on SO release date and monitor balance payments against agreed terms, so that I can ensure timely revenue recognition and proactively address overdue payments. Acceptance Criteria • System automatically tracks when a Sales Order (SO) is released by Sales Director.

			• Down payment schedule is created based on SO release date and payment terms. • Balance payment milestones are automatically tracked against due dates. • Finance Staff can query in chat (e.g., “<i>Show me outstanding balances for Customer X</i>”). • System generates proactive alerts in chat for: ○ Down payment not received within X days of SO release. ○ Balance payment overdue (D+1). ○ Upcoming payment due (D-7 reminder). • Chat displays customer, order ID, due amount, and status (Paid, Pending, Overdue). Design / System Enhancements • Payment milestone tracker integrated with MS5 financial data. • Alert rules engine for due/overdue reminders. • Chat integration for proactive notifications + on-demand enquiries. Chat interface provides summarized views of all active customer payment statuses, with the ability to request details on-demand (e.g., ‘Show me overdue payments for Customer X’).” Testing & Validation • Unit tests for milestone calculation logic (SO release → payment schedule). • Integration tests with MS5 for payment status sync. • UAT with Finance team to validate reminder timing, accuracy, and escalation flow. Technical Approach • MS5 API to retrieve payment schedules and statuses. • Event listener for SO release triggers. • Scheduled jobs to check overdue/approaching payments daily. • Chat API endpoints: ○ /api/chat/finance/payments/:customerId → query outstanding payments.
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			○ /api/chat/finance/alerts → push reminders to Finance Staff. Subtasks • Implement SO release → payment milestone mapping logic. • Build monitoring module for payment statuses. • Configure reminder rules (D-7, D+1). • Integrate with chat UI for queries + proactive alerts. • Logging for Finance audit trail. Dependencies • MS5 APIs for order + payment status. • Notification service (chat + email if escalation required). Risk Assessment • High: Missed overdue payment alerts → mitigation: daily batch check + manual override option. • Medium: Wrong due date calculation → mitigation: validate against finance contract terms. • Low: Too many notifications → mitigation: allow Finance to configure thresholds/frequency.
		Story 4.3: Escalate Overdue Payments	As a Finance Operations Specialist, I want the system to automatically escalate overdue down payments or balance payments to Sales Managers and Operations Leads, so that they are aware of customer payment issues and can follow up proactively to reduce risk exposure. Acceptance Criteria • When a down payment or balance payment becomes overdue (D+1), the system generates an escalation notification. • Escalations are routed automatically to: ○ Sales Manager of the account. ○ Operations Lead responsible for the order/project. • Chat interface displays escalation alerts with key details: ○ Customer Name, SO/PO reference, Amount Due, Days Overdue.

			• Sales/Operations can acknowledge the alert in chat and mark follow-up actions. • System supports “reminder escalation” if overdue > 7 days (e.g., notify Director-level). • Escalation history is logged for audit trail and dispute management. Design / System Enhancements • Extend finance monitoring module (from Story 4.2) with escalation rules. • Notification templates tailored for Sales vs Operations roles. • Chat integration for escalation alerts + quick actions (e.g., “<i>Mark as customer followed-up</i>”). • Chat interface supports drill-down queries to display escalated payment cases by status (e.g., ‘List all overdue escalations for ACME’ → shows Pending / In Progress / Resolved).” Testing & Validation • Unit tests for escalation rules (trigger on D+1, repeat on D+7). • Integration tests for routing alerts to correct Sales Manager/Operations Lead. • UAT with Finance, Sales, and Ops teams to validate escalation content and workflow. Technical Approach • Extend scheduled job from 4.2 to also trigger escalation events. • Role-mapping service to link customers → Sales Manager → Ops Lead. • Chat API endpoints: ◦ /api/chat/escalations → push escalation alerts. ◦ /api/chat/escalations/:id/acknowledge → mark action taken. • Store escalation log in audit DB with timestamps and acknowledgements. Subtasks • Define escalation thresholds (D+1, D+7). • Build escalation notification templates. • Integrate role-based routing (Sales/Operations). • Implement chat acknowledgement workflow. • Create escalation history log for reporting. Dependencies
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			• Payment monitoring logic from Story 4.2. • Notification and chat infrastructure. • Role/ownership mapping service. Risk Assessment • High: Escalation to wrong owner → mitigation: verify ownership mapping from CRM. • Medium: Alert fatigue if multiple overdue payments → mitigation: batch notifications per customer/account. • Low: Missed escalation log updates → mitigation: enforce DB transaction logging.
Sprint 5 – PoV Rollout & End-to-End Validation	Jan 26 – Feb 6, 2026	Story 5.1: PoV Rollout & End-to-End Validation	As a Program Owner, I want to conduct a Proof of Value (PoV) rollout across sales department, so that stakeholders can validate the end-to-end sales-to-finance workflow and confirm readiness for broader deployment. Acceptance Criteria • End-to-end flow from Opportunity Enquiry → Order Validation → Finance Monitoring runs live in a controlled PoV environment. • Key personas (Sales Manager, Sales Ops, Finance Staff) successfully execute core workflows via chat. • UAT feedback collected, logged, and prioritized for fixes. • Go/No-Go criteria for full rollout defined and documented. • Demo session delivered to stakeholders with walk-through of PoV use cases. Design / System Enhancements • Configure PoV environment with connections to MS5, IPAS, and CEC using controlled datasets. • Enable chat UI with full sprint 1–4 features. • Implement monitoring dashboards to track usage, issues, and adoption during PoV. Testing & Validation • End-to-end integration tests across Sales → Ops → Finance flows. • UAT sessions with selected users, covering success and failure paths.

			• Performance validation: ensure <3s response times under concurrent usage. • Feedback loop established (tickets, retrospectives). Technical Approach • Deploy consolidated backend with connectors (MS5, IPAS, CEC) into PoV environment. • Configure chat interface with role-based access (Sales Manager, Sales Ops, Finance). • Logging and monitoring tools enabled for tracking adoption. • Conduct stakeholder walkthrough and feedback sessions. Subtasks • Deploy PoV environment with sprint 1–4 features. • Set up monitoring dashboards for usage/adoption. • Run UAT sessions with Sales, Ops, Finance representatives. • Capture and prioritize feedback into backlog. • Prepare demo materials for stakeholder presentation. Dependencies • Completion of Stories 1.1–4.4. • Availability of representative datasets in MS5, IPAS, and CEC. • UAT participation from Sales, Ops, and Finance teams. Risk Assessment • High: PoV users may encounter blockers that delay validation → mitigation: provide dedicated support team during PoV. • Medium: Incomplete data in MS5/IPAS/CEC may limit test coverage → mitigation: preload representative sample data. • Low: Resistance to adoption in first demo → mitigation: prepare clear success metrics and demo script. As a Program Owner, I want to conduct a Proof of Value (PoV) rollout across selected business units, so that stakeholders can validate the end-to-end sales-to-finance workflow and confirm readiness for broader deployment.
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			Acceptance Criteria • End-to-end flow from Opportunity Enquiry → Order Validation → Finance Monitoring runs live in a controlled PoV environment. • Key personas (Sales Manager, Sales Ops, Finance Staff) successfully execute core workflows via chat. • UAT feedback collected, logged, and prioritized for fixes. • Go/No-Go criteria for full rollout defined and documented. • Demo session delivered to stakeholders with walk-through of PoV use cases. Design / System Enhancements • Configure PoV environment with connections to MS5, IPAS, and CEC using controlled datasets. • Enable chat UI with full sprint 1–4 features. • Implement logging and monitoring tools to track usage, issues, and adoption during PoV (internal monitoring; users interact only via chat). Testing & Validation • End-to-end integration tests across Sales → Ops → Finance flows. • UAT sessions with selected users, covering success and failure paths. • Performance validation: ensure <3s response times under concurrent usage. • Feedback loop established (tickets, retrospectives). Technical Approach • Deploy consolidated backend with connectors (MS5, IPAS, CEC) into PoV environment. • Configure chat interface with role-based access (Sales Manager, Sales Ops, Finance). • Enable logging and monitoring services for internal adoption tracking. • Conduct stakeholder walkthrough and feedback sessions. Subtasks • Deploy PoV environment with sprint 1–4 features. • Set up monitoring logs and usage/adoption reports. • Run UAT sessions with Sales, Ops, Finance representatives. • Capture and prioritize feedback into backlog.
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			• Prepare demo materials for stakeholder presentation. Dependencies • Completion of Stories 1.1–4.3. • Availability of representative datasets in MS5, IPAS, and CEC. • UAT participation from Sales, Ops, and Finance teams. Risk Assessment • High: PoV users may encounter blockers that delay validation → mitigation: provide dedicated support team during PoV. • Medium: Incomplete data in MS5/IPAS/CEC may limit test coverage → mitigation: preload representative sample data. • Medium: Resistance to adoption in first demo → mitigation: prepare clear success metrics and demo script.
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